

MD. NAZMUL ISLAM

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EDUCATION

- **Master of Science in Medicine** April 2025 – Present
Department of Medicine
Bangladesh Agricultural University, Mymensingh, Bangladesh.
CGPA: 3.80/4.00 (1st Semester)
Thesis: Integrating bacteriological culture with Bayesian Mixture Modeling to determine optimal somatic cell count thresholds for subclinical mastitis diagnosis in dairy cows.
- **Doctor of Veterinary Medicine (DVM)** Jan 2019 – Jan 2025
Faculty of Veterinary Science
Bangladesh Agricultural University, Mymensingh, Bangladesh.
CGPA: 3.487 out of 4.00

EXPERIENCE

- **Research Assistant** – Epidemiology and Preventive Medicine Laboratory, Department of Medicine, Bangladesh Agricultural University January 2023 – Present
 - Led field and lab investigations on infectious diseases in ruminants, contributing to multiple peer-reviewed publications.
 - Designed and executed ELISA, PCR, and DNA extraction protocols; analyzed data using R for Bayesian and multivariable modeling.
 - Generated publication-quality figures and visuals with Illustrator and Photoshop.
 - Co-authored manuscripts as first or equally contributing author, translating complex epidemiological data into clear scientific narratives.
 - Mentored junior researchers in study design, data collection, and lab procedures.

PUBLICATION

(Published)

- Alam, M.S., **Islam, M.N.**, Adhikari, B.J., Islam, S., Mahmud, R.S., Rahman, M.S., Ariful Islam, M., Aktaruzzaman, M., & Rahman, A.K.M.A. (2026).
Risk factors for latent and active brucellosis in dairy cattle: a Bayesian mixed-effects multinomial regression analysis in large-scale Bangladeshi herds. Animal Diseases, 6, 26.
<https://doi.org/10.1186/s44149-026-00234-w>
Contributed to sample collection and laboratory analysis, data curation, statistical analyses, Bayesian mixed-effects multinomial modelling, and manuscript drafting.
- Kibria, A.M., **Islam, M.N.**, Alam, M.S., Adhikari, B.J., Islam, S., Rafeh, A.A., Hasan, R.S.M., Rahman, M.S., & Rahman, A.K.M.A. (2026).
Estimated true prevalence and associated risk factors of bovine paratuberculosis antibodies in dairy herds in Bangladesh. PLOS ONE.
<https://doi.org/10.1371/journal.pone.0334683>
Equally contributing first author — contributed to study design, methodology development, Bayesian latent-class prevalence estimation, data analysis, visualization, and manuscript preparation.
- Mahmud Sajeeb, M.S., Alam, M.S., **Islam, M.N.**, Islam, M.M., Adhikari, B.J., Islam, S., Rahman, M.S., & Rahman, A.K.M.A. (2025).
Prevalence and risk factors of bovine viral diarrhea virus antibodies in dairy herds of Bangladesh. Veterinary Sciences, 12(8), 739.
<https://doi.org/10.3390/vetsci12080739>
Equally contributing first author — study design, field sampling, ELISA diagnostics, multivariable regression analysis, visualization, and manuscript preparation.

(Under review)

- **Islam, M.N.**, Haque, M.Z., Alam, M.S., Islam, S., Adhikari, B.J., Rahman, M.S., & Rahman, A.K.M.A. (2026).
Bayesian estimation of the true prevalence of Coxiella burnetii milk antibodies and associated risk factors in dairy

herds in Bangladesh. Veterinary Medicine and Science.

First author — led study design, Bayesian analysis, data interpretation, visualization, and manuscript preparation.

- Ferdous, M.J., **Islam, M.N.**, Alam, M.S., Saif, M.A., Rahman, M.A., Aktaruzzaman, M., Rahman, M.S., & Rahman, A.K.M.A. (2026).

Age-stratified transmission heterogeneity of bovine brucellosis in large commercial dairy herds in Bangladesh: a hierarchical Bayesian catalytic analysis. Epidemiology & Infection. (Equally contributing first author).

- Anwarul Islam, **Islam, M.N.**, Ferdous, M.J., Alam, M.S., Rahman, M.A., Mahmud, R.S., Alam, M.M., & Rahman, A.K.M.A. (2026).

Economic burden and strategic control of ruminant fascioliasis in Bangladesh. Journal of Helminthology.

- Alam, M.S., **Islam, M.N.**, Adhikari, B.J., Islam, S., Mahmud, R.S., Rahman, M.S., Ariful Islam, M., Aktaruzzaman, M., Meletis, L., Kostoulas, P., & Rahman, A.K.M.A. (2026).

Bayesian hierarchical mixture modelling to derive probabilistic iELISA thresholds for bovine brucellosis in Bangladeshi dairy herds. PLOS ONE.

- Aktaruzzaman, M., Ariful Islam, M., Shanta, S.A., Tahomina Akter, M., Hossain, M.T., Alam, M.S., **Islam, M.N.**, Meletis, L., Kostoulas, P., & Rahman, A.K.M.A. (2025).

Bayesian Gaussian mixture modeling to determine somatic cell count thresholds for subclinical mastitis diagnosis in dairy cows. The Journal of Veterinary Medical Science.

- Aktaruzzaman, M., M. Tahomina Akter, Mazhar Islam, M., Ariful Islam, M., Hossain, M.T., Alam, M.S., **Islam, M.N.**, & Rahman, A.K.M.A. (2025).

Economic impact of subclinical mastitis in Bangladeshi crossbred dairy cattle. Preventive Veterinary Medicine.

- Aktaruzzaman, M., Ariful Islam, M., Shanta, S.A., Tahomina Akter, M., Hossain, M.T., **Islam, M.N.**, Alam, M.S., & Rahman, A.K.M.A. (2025).

Prevalence and cow-level risk factors for mild and severe subclinical mastitis in intensive and extensive dairy systems in Sirajganj, Bangladesh. Veterinary Research Communications.

- Saha, U.K., **Islam, M.N.**, Chaudhary, P., Hassan Alif, M.K., Mim, S.N., & Rahman, M.S. (2026).

Exploratory longitudinal ELISA antibody responses and hematological parameters after heat-inactivated Brucella abortus vaccination in cattle. Bulgarian Journal of Veterinary Medicine. (Equally contributing first author)

- Saif, M.A., **Islam, M.N.**, Ferdous, M.J., Rahman, A.K.M.A., & Bari, F.Y. (2026).

Early-life feeding and management interventions on lamb survivability. Bulgarian Journal of Veterinary Medicine.

- Saif, M.A., **Islam, M.N.**, Ferdous, M.J., Rahman, A.K.M.A., & Bari, F.Y. (2026).

Integrated immune and protein profiles predict growth in preweaning lambs independent of feeding regimen and maternal oxytocin administration. Iraqi Journal of Veterinary Sciences.

CONFERENCE PAPERS & PROCEEDINGS

- Islam, M.; Alam, M.; Islam, S.; Adhikari, B.; Rahman, A. **Feasibility of on-farm somatic cell count screening for subclinical mastitis: Prevalence and risk factors in a commercial dairy herd in Bangladesh**, in *Proceedings of the 1st International Online Conference on Veterinary Sciences, 3–5 December 2025, MDPI: Basel, Switzerland*, <https://sciforum.net/paper/view/27422>
- Alam, M. S., Islam, M. N., Rafeh, A. A., Adhikari, B. J., Kibria, M. A., Mahmud, M. S., Ema, A. A., Islam, S., Rahman, M. S., Rahman, A. K. M. A. (2025). **Herd-level prevalence of brucellosis, paratuberculosis, neosporosis, and bovine viral diarrhea in large dairy herds in Bangladesh.** *BAU Research Progress*, 35.
- Alam, M. S., Islam, M. N., Rafeh, A. A., Rahman, M. S., Islam, M. A., & Rahman, A. K. M. (2024). **Herd-level prevalence of brucellosis in large dairy herds in Bangladesh.** *Proceedings of the BSVER Annual Scientific Conference 30*, Bangladesh Agricultural University, Mymensingh, February 24-25, 2024.

CONFERENCE PRESENTATIONS

- Feasibility of On-Farm Somatic Cell Count Screening for Subclinical Mastitis: Prevalence and Risk Factors in a Commercial Dairy Herd in Bangladesh.** Presented at IOCVS 2025, 4 December 2025. DOI: 10.13140/RG.2.2.16208.16642.
- Prevalence and Epidemiology of Bovine Brucellosis in Large Dairy Farms of Bangladesh.** Presented a research poster on the prevalence, risk factors, and control strategies of bovine brucellosis at the BSVER Annual Scientific Conference 2024, Bangladesh Agricultural University, Mymensingh, February 24-25, 2024.

TECHNICAL SKILLS

- **Laboratory Techniques:** Indirect ELISA, DNA extraction, PCR, microscopy, histopathology
- **Statistical Analysis & Programming:** R (data analysis, visualization, and statistical modeling)
- **Geospatial Mapping & Analysis:** QGIS (used for spatial data visualization and disease mapping)
- **Scientific Visualization & Design:** Adobe Illustrator, Adobe Photoshop, Canva, BioRender (creating research figures, infographics, posters, and dynamic presentations)
- **Productivity Software:** Microsoft Office Suite (Word, Excel, PowerPoint for report writing and presentations)

LINGUISTIC PROFICIENCY

- English (Fluent working proficiency; medium of instruction throughout academic career)

TRAINING & WORKSHOP

- **Veterinary Intern** **July 2023 – December 2023**
Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh
 - Conducted clinical assessments, treatments, and surgical assistance at notable institutions.
 - Strengthened veterinary skills across clinical, farm, and wildlife settings under expert guidance.
- **Agricultural Extension Education (6-day Field Trip)** **Jan 2024 – Jan 2024**
Department of Agricultural Extension Education, Bangladesh Agricultural University, Mymensingh

LEADERSHIP & PROFESSIONAL DEVELOPMENT

- **Aspire Leadership Program, Finalist & Alumni** **Program completed in August 2025**
Selected as one of 6,175 finalists from over 36,000 applicants to complete the competitive 2-month 2025 Aspire Leaders Program. Completed 40 hours of intensive coursework developed by Aspire Institute, founded by Harvard Business School professors. Gained advanced leadership, critical thinking, and communication skills, with a strong focus on creating global social impact.

MEMBERSHIP & AWARDS

- American Society for Microbiology (Global Outreach) – Student Member **June 2024 – Present**
- Badhon (Voluntary Blood Donor's Organization) – Member **Feb 2019 – Present**
- Merit Scholarship awarded by Education Board for Secondary School Certificate (SSC), 2015 and Higher Secondary Certificate (HSC), 2017.
- **National Science and Technology (NST) Fellowship**, Ministry of Science and Technology, Government of Bangladesh, 2026
Awarded in support of Master's thesis research in veterinary medicine and animal health, in recognition of academic merit and research potential.

CERTIFICATIONS

- R Programming for Research – Udemy, July 2024.
- Tropical Parasitology: Protozoans, Worms, Vectors and Human Diseases – Coursera, May 2020.
- Animal Behavior and Welfare – Coursera, April 2020.

REFERENCES

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